Software Project Lab - 3 Project Proposal

APEX Net

Anatomically **P**rior-**E**nhanced **X**-Ray **Net**work for Thoracic Disease Classification and Localisation

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Brief

Project Overview

APEX-Net is an AI-powered system that analyses chest X-ray images to automatically detect different thoracic diseases and highlight the affected areas.

Assist Doctors in Underserved Areas

Catch Rare Diseases Early

Make AI Decisions Transparent

THE PROBLEM

Three Critical Gaps in Existing AI Systems

1

Rare Disease Blindness

2

No Anatomical Focus

3

No Visual Explanation

PROPOSED SOLUTION

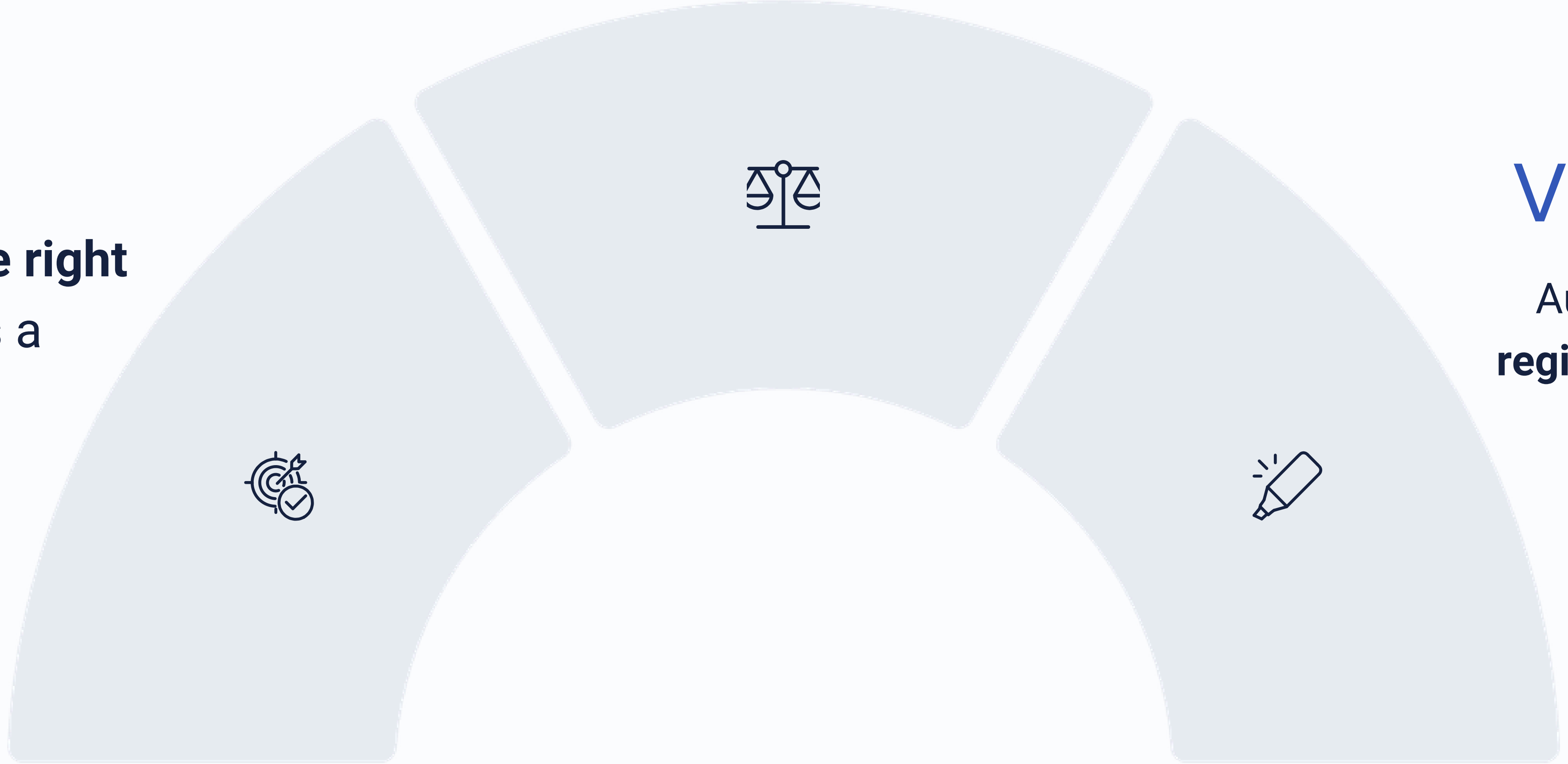
APEX-Net: One System, Three Answers

Balanced Learning

A specialised training strategy that gives **extra emphasis to rare diseases**, reducing missed diagnoses.

Smart Attention

Guides the model to focus on **the right anatomical regions** — just as a radiologist would.

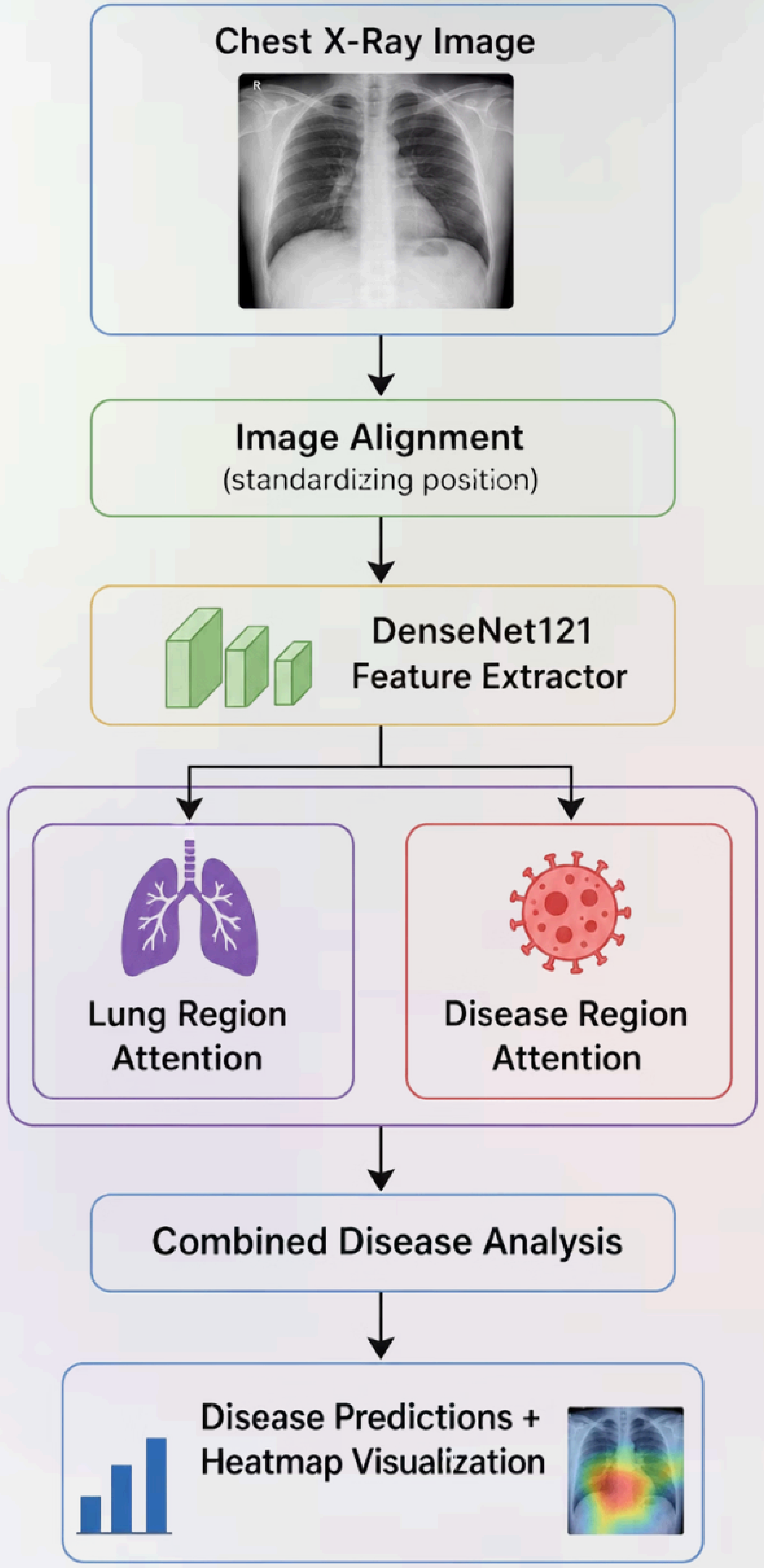


Visual Explanation

Automatically **highlights the affected region** in the X-ray so clinicians can verify every prediction.

METHODOLOGY

How APEX-Net Works



EXPECTED OUTCOMES

What This Project Will Deliver

Disease Detection

Accurately classify **14 distinct chest diseases** from a single X-ray image.

Localisation

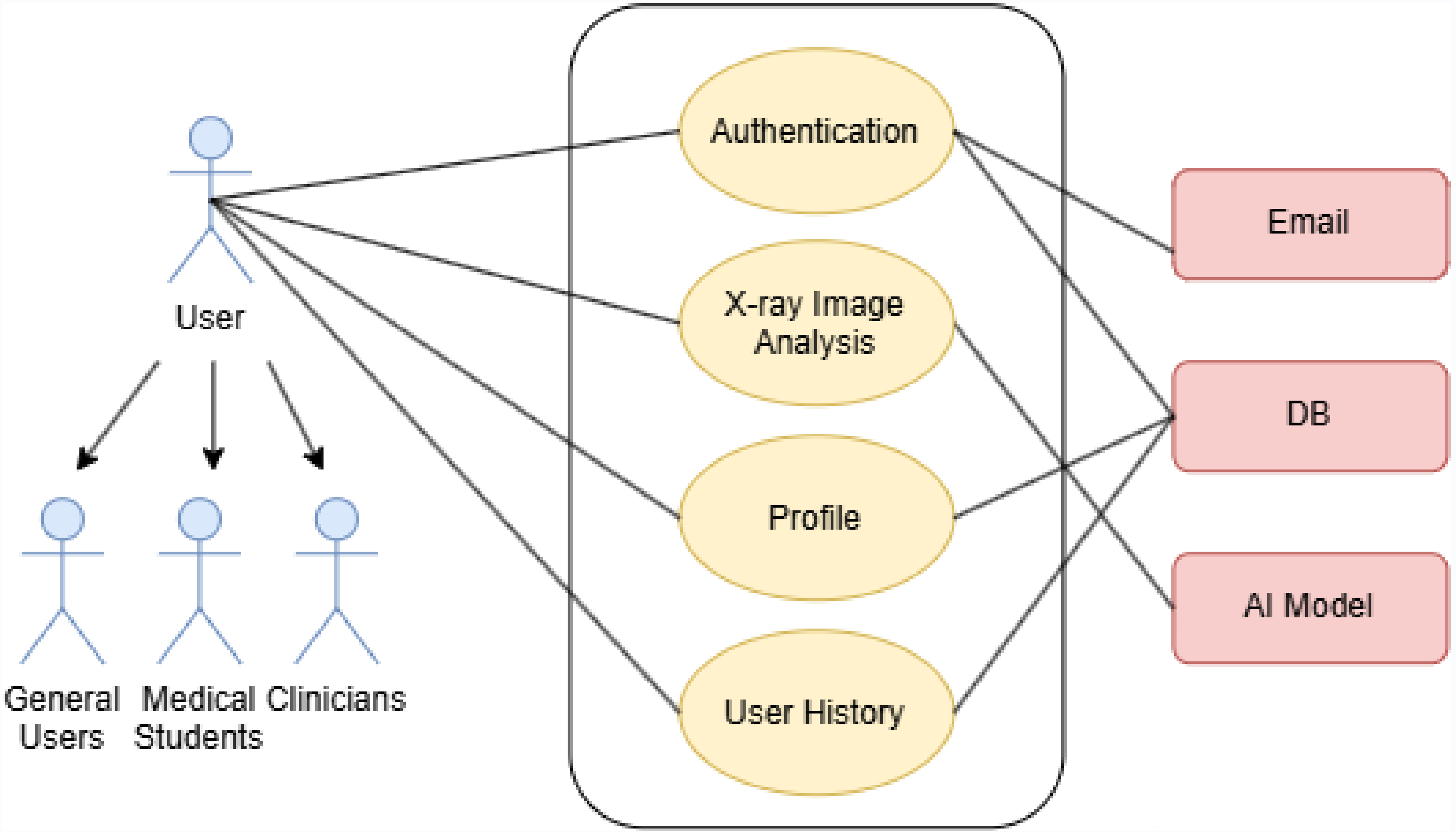
Highlight **exactly where** in the X-ray each detected disease is located.

Web Application

A working **demonstration interface** that clinicians or reviewers can interact with directly.

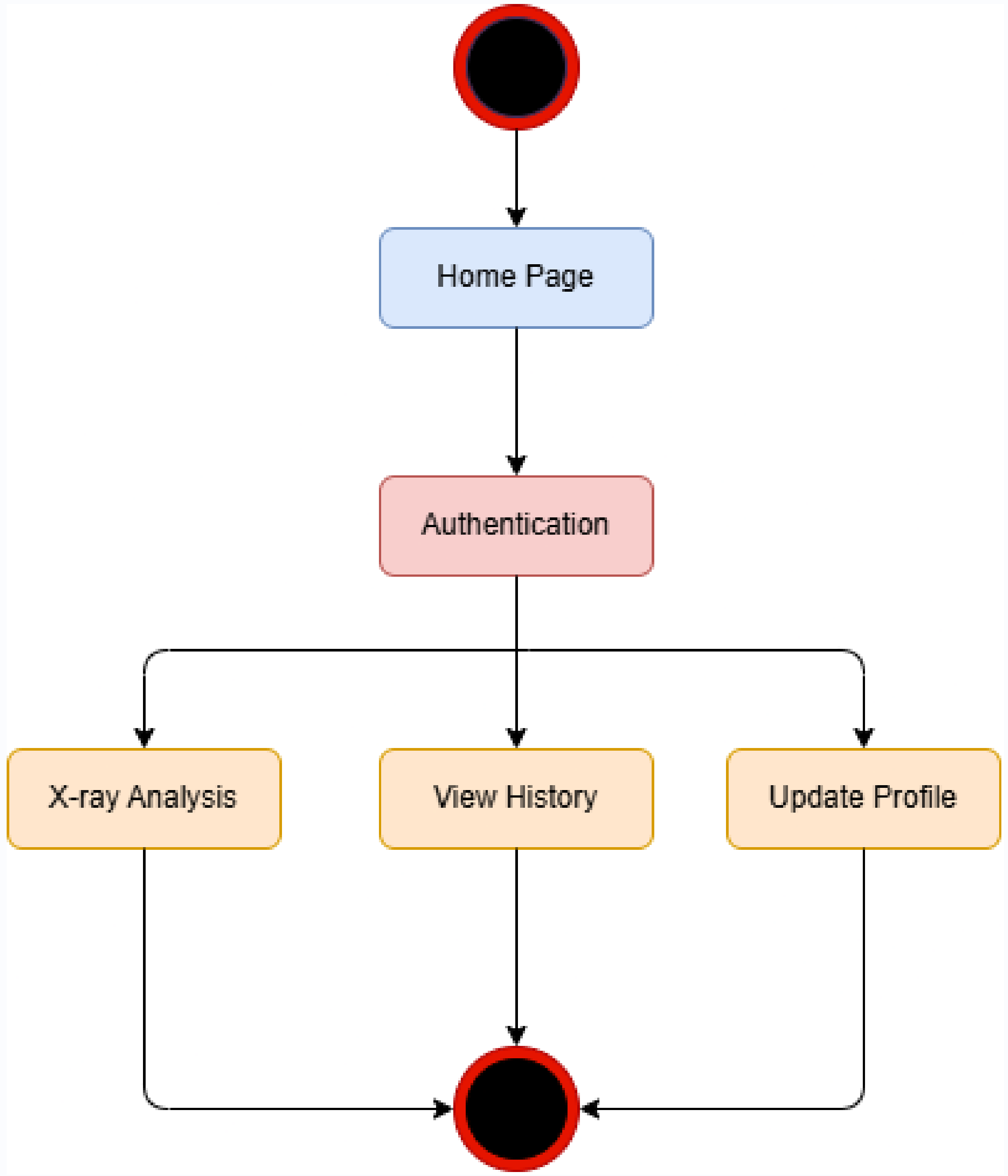
Diagrams

Use Case level 1.0



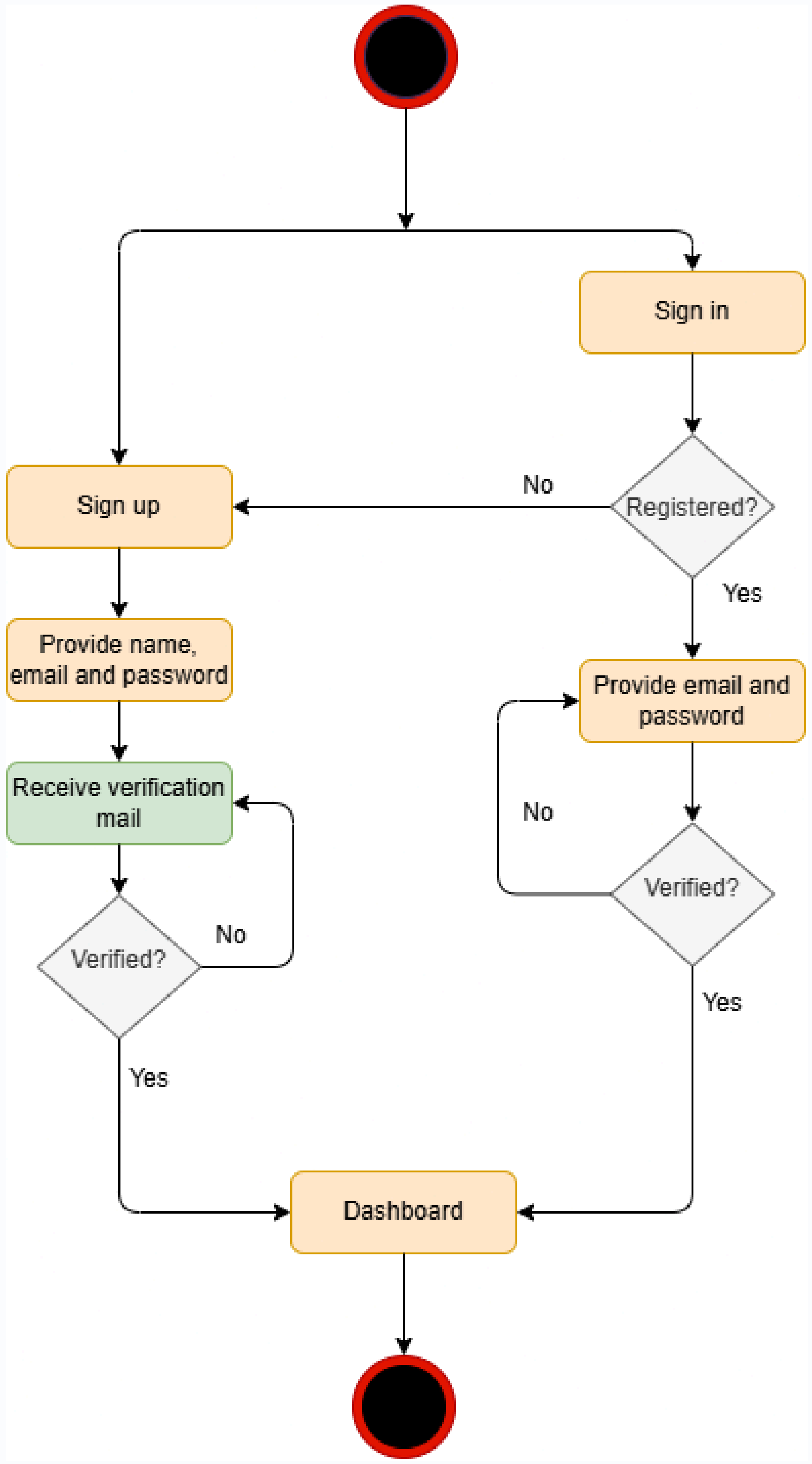
Diagrams

Activity level 1.0



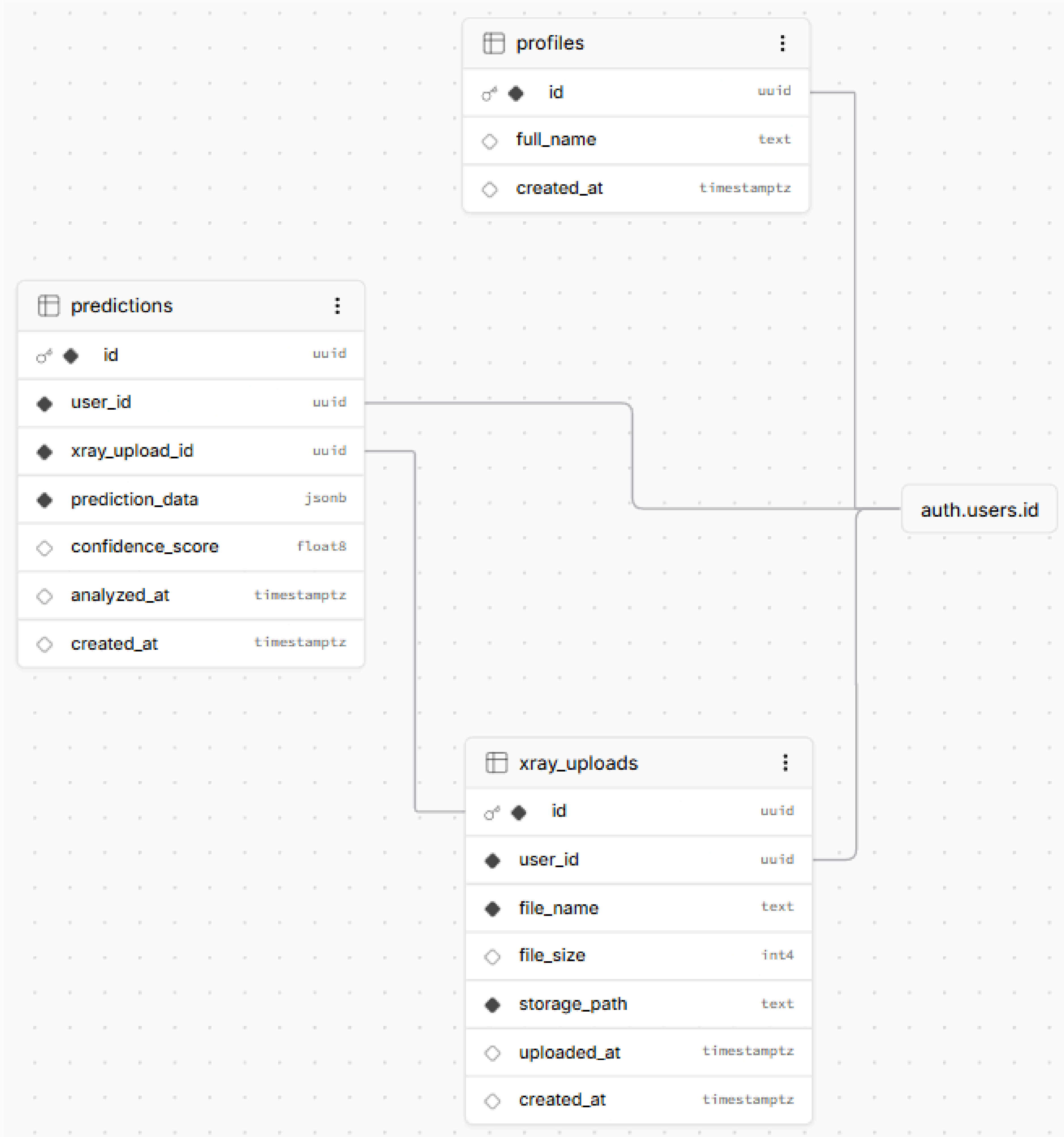
Diagrams

Activity level 1.1 -Authentication



Diagrams

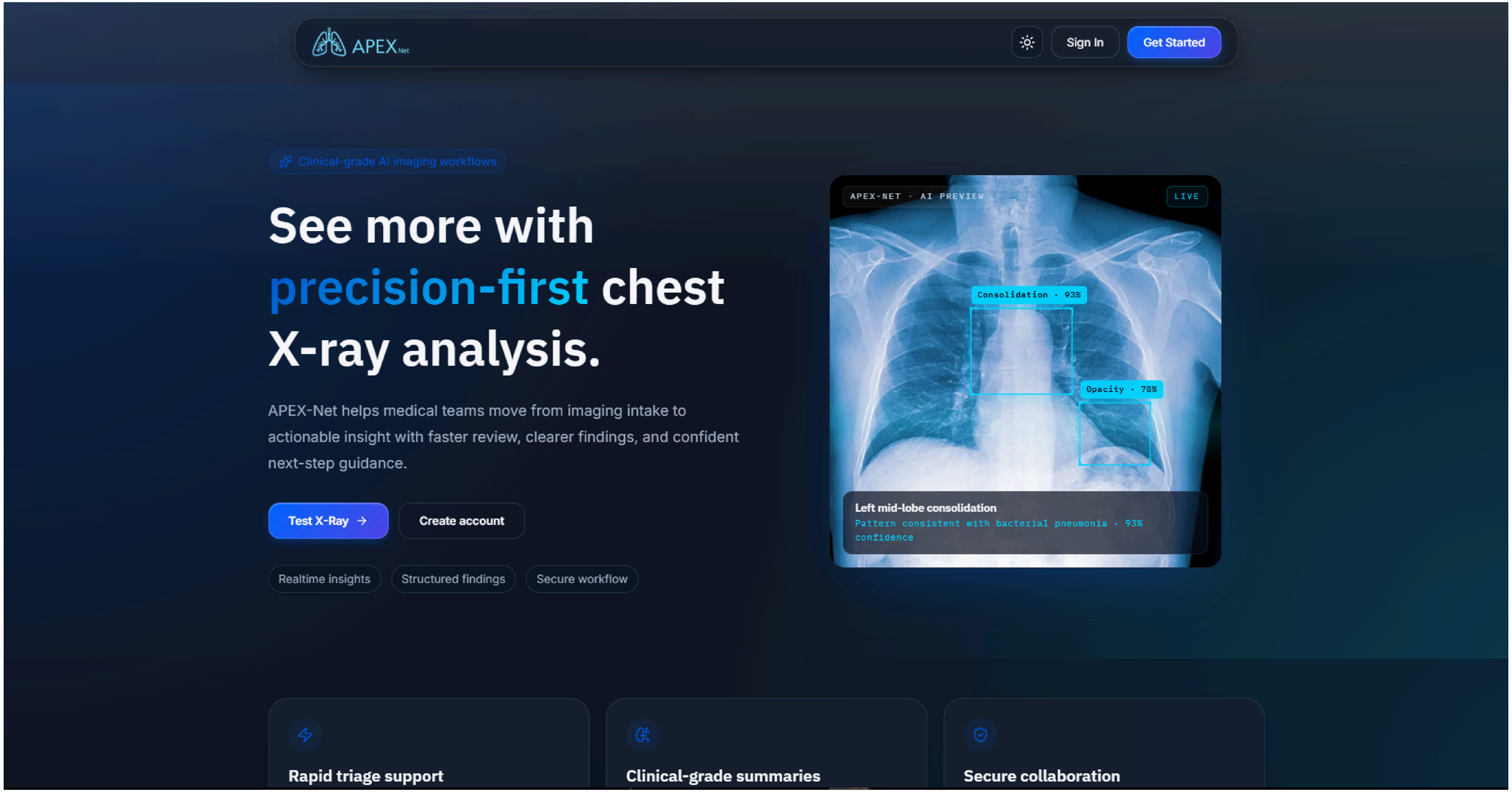
ER



profiles
id uuid
full_name text
created_at timestamptz
predictions
id uuid
user_id uuid
xray_upload_id uuid
prediction_data jsonb
confidence_score float8
analyzed_at timestamptz
created_at timestamptz
auth.users.id
xray_uploads
id uuid
user_id uuid
file_name text
file_size int4
storage_path text
uploaded_at timestamptz
created_at timestamptz

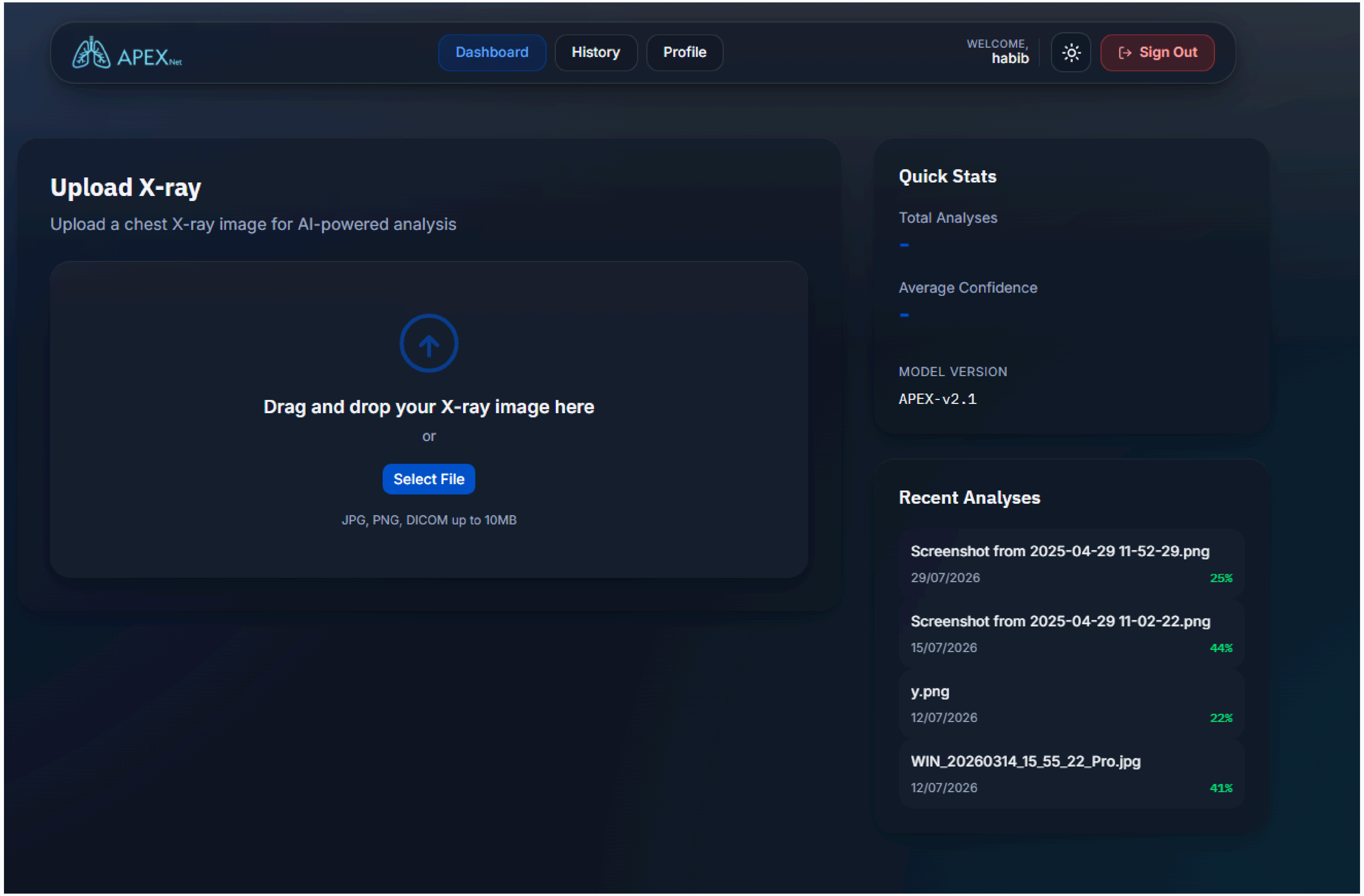
Demo SS

UI


APEX Net
Sign In
Get Started
Clinical-grade AI imaging workflows
See more with precision-first chest X-ray analysis.
APEX-Net helps medical teams move from imaging intake to actionable insight with faster review, clearer findings, and confident next-step guidance.
Test X-Ray →
Create account
Realtime insights
Structured findings
Secure workflow
APEX-NET · AI PREVIEW
LIVE
Consolidation · 93%
Opacity · 78%
Left mid-lobe consolidation
Pattern consistent with bacterial pneumonia · 93% confidence
Rapid triage support
Clinical-grade summaries
Secure collaboration

Demo SS

UI



APEXNet
Dashboard
History
Profile
WELCOME,
habib
Sign Out
Upload X-ray
Upload a chest X-ray image for AI-powered analysis
Drag and drop your X-ray image here
or
Select File
JPG, PNG, DICOM up to 10MB
Quick Stats
Total Analyses
-
Average Confidence
-
MODEL VERSION
APEX-v2.1
Recent Analyses
Screenshot from 2025-04-29 11-52-29.png
29/07/2026
25%
Screenshot from 2025-04-29 11-02-22.png
15/07/2026
44%
y.png
12/07/2026
22%
WIN_20260314_15_55_22_Pro.jpg
12/07/2026
41%

Progress

Model Training

```
(apex_env) PS E:\Projects\APEX-Net\APEX-Net\DacNet> python scripts/dacnet.py --data_dir D:\NIH_data
Using dataset directory: D:\NIH_data
wandb: Loading settings from E:\Projects\APEX-Net\APEX-Net\DacNet\wandb\settings
wandb: Tracking run with wandb version 0.28.1
wandb: W&B syncing is set to `offline` in this directory. Run `wandb online` or set WANDB_MODE=online to enable cloud syncing.
wandb: Run data is saved locally in E:\Projects\APEX-Net\APEX-Net\DacNet\wandb\offline-run-20260801_213255-bspvyjbg
wandb: View this run in the terminal with `wandb leet`
Epoch 1/25 [Train]:  13%|█████                                    | 1646/12981 [30:53<3:37:01,  1.15s/it, loss=0.0523]
```

Thank You